

Table Of Contents

About this book	xi
1. Power Platform Overview	1
1.1 Pillars.....	1
1.2 Licensing.....	5
1.3 Platform Components	9
1.4 Historical Background.....	12
1.5 Summary.....	21
2. Architecture.....	22
2.1 Authentication and Authorization	22
2.1.1 App registration using Azure Portal	22
2.1.2 App registration using PAC CLI	27
2.2 Communication Protocols.....	30
2.2.1 Web API.....	30
2.2.2 Tabular Data Stream	35
2.3 Business Rules	39
2.4 Client-side scripting.....	41
2.4.1 Web Resources	41
2.4.2 PCF Controls	52
2.5 Classic Workflows	62
2.6 Power Automate	68
2.6.1 Cloud Flows and Desktop Flows.....	68
2.6.2 Connectors.....	71
2.6.3 Connectors Architecture	73
2.6.4 Connections - Security	75
2.7 Plugins	87
2.7.1 Internals	87
2.7.2 Virtual Entities	91

2.7.3 Early Bound vs Late Bound Entities	97
2.7.4 Managed Identities	98
2.8 Dataflows	107
2.8.1 Standard Dataflows	107
2.8.2 Analytical Dataflow	108
2.8.3 Data Import.....	108
2.9 Networking and Data Gateways	111
2.9.1 Microsoft On Premise Data Gateway.....	111
2.9.2 Subnet Delegation	112
2.10 Summary	115
3. Integration Aspects	117
3.1 Security.....	117
3.1.1 Power Platform Firewall	118
3.1.2 Authentication	118
3.1.3 Authorization	118
3.1.4 Network Layer Security Controls	120
3.2 Auditing.....	121
3.3 Multiplexing.....	125
3.4 Regulatory and Compliance	128
3.5 Performance.....	131
3.5.1 Indexes	133
3.5.2 Alternate Key.....	133
3.5.3 Quick Find Indexes	137
3.5.4 Data Model	137
3.5.5 ServiceClient Configuration	142
3.5.6 Plugins.....	150
3.5.7 Security Roles and POA	151
3.5.8 Bypassing Dataverse Background Processing.....	152
3.6 High Availability.....	154
3.7 Reliability and Resilience	157

3.7.1 ServiceClient	157
3.7.2 Plugins	161
3.7.3 Error Handling in Power Automate	163
3.7.4 Optimizations in Power Automate	168
3.8 Telemetry Data Collection	174
3.8.1 Plugin Trace Logs.....	174
3.8.2 Application Insights.....	177
3.9 Monitoring	185
3.9.1 Power Automate Monitoring.....	186
3.10 Observability	191
3.11 Summary	193
4. Azure Services	194
4.1 Azure Functions	194
4.1.1 Webhook registration	195
4.1.2 Custom Connectors	197
4.1.3 Custom Plugins.....	198
4.1.4 Use Cases For Azure Functions	198
4.1.5 Summary.....	200
4.2 Azure Service Bus	202
4.2.1 Point to Point	202
4.2.2 Topics and subscriptions (Pub/Sub).....	204
4.3 Azure Event Hub	208
4.4 Azure API Management	212
4.4.1 Custom Connector For API	214
4.4.2 Pro code – Client from OpenAPI definition	218
4.5 Azure Logic Apps.....	222
4.5.1 Logic Apps vs Power Automate.....	222
4.5.2 Hosting Plans.....	226
4.5.3 Built-in vs Managed Connectors.....	227
4.6 Azure Data Factory.....	230

4.6.1 Data Factory vs. Dataflows	231
4.6.2 Use Cases in the Dataverse ecosystem.....	237
4.6.3 Optimizations and performance tips.....	240
4.7 Account Storage	243
4.7.1 Integration examples	244
4.7.2 Power Platform File Storage vs. Azure Account Storage	245
4.7.3 Use cases in the Dataverse ecosystem	247
4.8 Summary.....	249
5. Real World Examples	253
5.1 Migrating from on premise to cloud	253
5.1.1 Hybrid Deployment	253
5.1.2 As Is Architecture	257
5.1.3 Requirements	259
5.1.4 To Be Architecture	260
5.1.5 Implementation plan	261
5.2 Client Side - IFrame integration.....	269
5.3 Communication widgets	272
5.3.1 Channel Integration Framework (CIF).....	272
5.3.2 Generic Sessions vs Entity Sessions	274
5.4 Acting on behalf signed in users.....	281
5.5 Virtual entities	291
Index	294